

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular End Semester Examination – Summer 2022

Course: B. Tech.

Branch: E & TC

Semester: VI

Subject Code & Name: BTETPE604A CMOS Design

Max Marks: 60

Date: 26/08/2022

Duration: 3.45 Hr.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q.1 Solve the following

- A) Draw neat diagrams and explain CMOS n-well fabrication process in detail. 2 12

Q.2 Solve the following

- A) Draw schematic & stick diagram for two input CMOS XOR gate with equal rise and fall time. 2 4
- B) Draw Layout for two input static CMOS XOR gate 2 8

Q. 3 Solve Any Two of the following.

- A) Explain Channel Length Modulation (CLM) the non-ideal effect in MOSFET. 1 6
- B) Implement logic function $Y = \overline{(A + B + C)} \cdot \overline{(D + E)}$ using Dual rail Domino CMOS logic 2 6
- C) Explain static and dynamic power in CMOS circuits. What is an activity factor? 1 6

Q.4 Solve Any Two of the following.

- A) Draw Schematic, stick diagram and layout for two input CMOS NOR gate with equal rise and fall time. Use Static CMOS design style. 2 6
- B) What are skewed gates? Draw high skewed and low skewed two input static CMOS XNOR gate. 2 6
- C) Implement JK Flip-Flop using Static CMOS design style. 2 6

Q. 5 Solve Any Two of the following.

- A) Explain in detail CMOS λ -based design rules. 1 6
- B) Draw and explain the Voltage Transfer Characteristics for CMOS inverter in detail. 1 6
- C) For implementation of 4 input NAND gate, compare various CMOS logic design styles based on, No. of transistors, ease of complexity, delay, Power. 1 6

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